SHARAN KALAMOHAN

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EDUCATION

MSc Automotive & Motorsport Engineering — Brunel University London Sep 2025 – Sep 2026
Predicted Distinction

- **Modules:** Vehicle Dynamics & CAD (87%); Vehicle Propulsion Technologies (71%); Racing Vehicle Design (70%).
- **Dissertation (Due Sep 2026):** Air intake optimisation for a hydrogen-converted high-performance engine, using coupled 1D–3D modelling in Ricardo WAVE and Ansys Fluent to maximise volumetric efficiency.
- **Coursework:** P0/P1/P2 hybrid architecture study in Simulink Powertrain Blockset (73%); MATLAB Formula 4 suspension optimisation (83%); Rally-stage telemetry and DC motor control (78%).

BEng (Hons) Mechanical Engineering — University of Nottingham Sep 2021 – Jun 2025
2:1 (Upper Second Class Honours), including a 12-month industrial placement at Renishaw plc

- **Modules:** Powertrain Engineering (91%); Automotive Technology (71%).
- **Dissertation (71%):**
  - Validated an Abaqus CAE torsional FEA of concentric DCT input shafts (420 Nm) against classical torsion theory to within 2.2–7.1% on far-field shear stress, resolving a 3.47 spline-root stress concentration factor.
  - Ranked 13 steel grades with a weighted Materials Performance Index; 50B46 tripled inner-shaft factor of safety (2.04 → 6.23) for +1.35% material cost, with no geometry change.

TECHNICAL SKILLS

- **CAD & PLM:** Siemens NX; SolidWorks; Autodesk Inventor; Teamcenter PLM; 3D scanning and reverse engineering.
- **Simulation:** MATLAB & Simulink; Ricardo WAVE; Ansys Fluent; Abaqus CAE; IPG Carmaker.
- **Manufacturing:** GibbsCAM; CNC milling and turning; CMM programming (Renishaw MODUS); VCDS data logging.

ENGINEERING EXPERIENCE

Mechanical Engineering Placement Student — Renishaw plc, Wotton-under-Edge Aug 2023 – Aug 2024
Gauging R&D — Equator X gauging system, since launched and named Renishaw's internal Product of the Year 2024.

- Designed and machined a magnetic multi-slot bonding fixture, raising carbon-fibre strut output 5x.
- Planned and ran structural-adhesive life, thermal expansion and bond-line test programmes, plus tensile testing of carbon-fibre struts; authored the internal test reports that drove adhesive selection and production feasibility
- Cut bond-line thickness spread from 800 µm to 50 µm, achieving the CTE stability for the calibration software.
- Modelled components in Siemens NX under Teamcenter PLM, programmed GibbsCAM toolpaths, machined the parts and verified them by CMM inspection, owning the loop from concept to measured part.

ENGINEERING PROJECTS

Engineering Technical Director — Brunel Masters Motorsport (Formula Student UK) Oct 2025 – May 2026
FSUK Concept Class; team of 30 across four engineering groups developing an electric Formula Student concept car.

- Technical authority across Powertrain, Chassis, Suspension and Driver Controls groups: chaired design reviews to resolve cross-subsystem clashes, ran weekly meetings and owned the master CAD assembly and specification.

- Set and arbitrated vehicle-level targets — 280 kg, 453 V nominal, 40:60 front/rear — that all four engineering groups designed to.
- Delivered the May Concept Class design submission: full vehicle CAD and specifications showing rule compliance.

Traction Motor Selection & Drivetrain Simulation — MSc Major Group Project Sep 2025 – May 2026

Individual technical contribution as Motor Subsystem Lead on the Formula Student concept car.

- Developed a forward–backward quasi-static lap simulation in MATLAB for all four FS dynamic events, integrating a digitised 6,271-point Emrax efficiency map and a lumped-parameter thermal model.
- Benchmarked predicted acceleration against the 2025 FSUK winning 75 m time to within 0.7%.
- Selected the Emrax 228 MV LC motor from five candidates against inverter, accumulator and coolant rule constraints, and optimised the final-drive ratio to 3.60 via a 144-run weighted sweep.
- Identified a 7 kW DC-bus rule breach at 80 kW shaft-power; implemented a torque cap to ensure compliance.
- Designed a state-of-charge-aware adaptive torque controller that recovered 94 s of simulated endurance time over the 22 km endurance event versus a fixed, derated torque map, while still guaranteeing accumulator range.
- Showed that regenerative recovery (7.1%) is limited by accumulator C-rate rather than motor capability, redirecting the team's next design priority.

1D Engine Modelling & Vehicle Data Correlation — Self-directed Jul 2026 – Present

- Logged full-throttle and part-load data from a turbocharged 2.0 TFSI engine via VCDS: boost pressure, mass air flow, intake air temperature, lambda and ignition timing.
- Building a 1D air-path model in Ricardo WAVE around scaled K04 compressor and turbine maps, correlating against the logged data and published dynamometer curves toward a hydrogen conversion feasibility study.

ADDITIONAL EXPERIENCE

Assistant Area Manager — Mohan Retail Ltd, Amersham Sep 2022 – Present (Flexible Part Time)

- Managed logistics, stock and staff scheduling across four family run retail stores alongside full-time study.
- Introduced Excel-based stock rotation and dynamic pricing, cutting spoilage and increasing weekly revenue 3.6%.

ADDITIONAL SKILLS & INTERESTS

- **Languages:** English (native); Tamil (native).
- **Memberships:** IMechE Student Member.
- **Interests:**
    - Music production: self-taught in FL Studio, working with synthesis, sampling and mixing.
    - Strength training through a programme balancing strength, nutrition and recovery.
    - Weekly 5-a-side football in a local competitive league.